

MST300 - Introduction to Microsoft Cloud Technologies

Fall 2026 Syllabus, Section NBB, Class Nbr 5219

Course Description

This course introduces student to Microsoft cloud services. In this course students will create cloud solutions using Microsoft Azure. Students will gain hands-on experience by deploying Azure services, core solutions and management tools and security features. Students will explore how identity and privacy, governance and compliance, cost management and service level agreements impact businesses through in-class activities.

Prerequisite(s)

MST200

Instructor Information

Nooshin Beheshti - Primary Instructor

Email: nooshin.beheshti@senecapolytechnic.ca

In-Person Mode

This class will be taught via an **in-person** instruction mode. Students are required to attend all classes on campus.

Times and Location

T 9:50am-11:35am in Newnham Bldg A - A4515

M 9:50am-11:35am in Newnham Bldg C - C3036

Course Learning Outcomes

Upon successful completion of this course the student will be able to:

1. Explain general cloud computing concepts to support cloud solutions within an organization.
2. Evaluate core services available with Microsoft Azure to recommend an appropriate service for a business scenario.
3. Analyze core solutions and management tools in Microsoft Azure to support business requirements.
4. Apply security and network security features within Microsoft Azure to support the business operations.
5. Support the organization by applying the appropriate identity, governance, privacy, and compliance capabilities of Microsoft Azure.
6. Analyze cost management and Service Level Agreements (SLAs) with Microsoft Azure to improve infrastructure support costs.

Essential Employability Skills

- respond to written, spoken, or visual messages in a manner that ensures effective communication
- execute mathematical operations accurately
- apply a systematic approach to solve problems
- use a variety of thinking skills to anticipate and solve problems
- locate, select, organize, and document information using appropriate technology and information systems
- analyse, evaluate, and apply relevant information from a variety of sources
- show respect for the diverse opinions, values, belief systems, and contributions of others
- interact with others in groups or teams in ways that contribute to effective working relationships and the achievement of goals
- manage the use of time and other resources to complete projects
- take responsibility for one's own actions, decisions, and consequences
- communicate clearly, concisely, and correctly in the written, spoken, and visual form that fulfils the purpose and meets the needs of the audience

Course Topics

MST300

- Core Azure Concepts
- Core Azure Services
- Core Solutions and Management Tools on Azure
- General Security and Network Security Features
- Identity, Governance, Privacy, and Compliance Features
- Azure Cost Management and Service Level Agreements

To find out the cost of books and learning material go here (<https://www.bkstr.com/senecastore/shop/textbooks-and-course-materials>).

Any courses not listed on the bookstore webpage do not require any resources for purchase. All resources will be provided by your instructor.

Required Supplies

Microsoft Azure Subscription

Modes of Evaluation

Assessment Type	Percentage
Management of budget allotment	10%
Labs (minimum 20)	20%
Quizzes (minimum 6)	15%
Projects (minimum 2)	30%
Test	25%
Total	100%

Note

To obtain a credit in this course, a student must:

1. Achieve a grade of 50% or higher in the overall course
2. Achieve a weighted average of 50% or higher on all labs
3. Achieve a weighted average of 50% or higher on all projects

Schedule of Topics and Assignments

Week of	Agenda/Topic	Reading(s)	Due
9/8	01 Describe core Azure concepts Identify the benefits and considerations of using cloud services: • identify the benefits of cloud computing, such as High Availability, Scalability, Elasticity, Agility, and Disaster Recovery • identify the differences between Capital Expenditure (CapEx) and Operational Expenditure (OpEx) • describe the consumption-based model Describe the differences between categories of cloud services: • describe the shared responsibility model • describe Infrastructure-as-a-Service (IaaS) • describe Platform-as-a-Service (PaaS) • describe serverless computing • describe Software-as-a-Service (SaaS) • identify a service type based on a use case 01 Describe core Azure concepts Describe the differences between types of cloud computing: • define cloud computing • describe Public cloud • describe Private cloud • describe Hybrid cloud • compare and contrast the three types of cloud computing	Module 1 Microsoft Learning Online Docs (Pls see "Weekly Readings" in Blackboard)	
9/15	02 Describe Core Azure Services Describe the core Azure architectural components: • describe the benefits and usage of Regions and Region Pairs • describe the benefits and usage of Availability Zones • describe the benefits and usage of Resource Groups • describe the benefits and usage of Subscriptions • describe the benefits and usage of Management Groups • describe the benefits and usage of Azure Resource Manager explain Azure resources	Module 2 Microsoft Learning Online Docs (Pls see "Weekly Readings" in Blackboard)	
9/22	02 Describe Core Azure Services Describe core resources available in Azure: • describe the benefits and usage of Virtual Machines, Azure App Services, Azure Container Instances (ACI), Azure Kubernetes Service (AKS), and Windows Virtual Desktop • describe the benefits and usage of Virtual Networks, VPN Gateway, Virtual Network peering, and ExpressRoute	Module 2 Microsoft Learning Online Docs (Pls see "Weekly Readings" in Blackboard)	Quiz 1 (2.5%)
9/29	02 Describe Core Azure Services Describe core resources available in Azure: • describe the benefits and usage of Container (Blob) Storage, Disk Storage, File Storage, and storage tiers • describe the benefits and usage of Cosmos DB, Azure SQL Database, Azure Database for MySQL, Azure Database for PostgreSQL, and SQL Managed Instance • describe the benefits and usage of Azure Marketplace Introduction to Project 1	Module 2 Microsoft Learning Online Docs (Pls see "Weekly Readings" in Blackboard)	

10/6	03 Describe core solutions and management tools on Azure Describe core solutions available in Azure: • describe the benefits and usage of Internet of Things (IoT) Hub, IoT Central, and Azure Sphere • describe the benefits and usage of Azure Synapse Analytics, HDInsight, and Azure Databricks • describe the benefits and usage of Azure Machine Learning, Cognitive Services and Azure Bot Service • describe the benefits and usage of serverless computing solutions that include Azure Functions and Logic Apps • describe the benefits and usage of Azure DevOps, GitHub, GitHub Actions, and Azure DevTest Labs	Module 3 Microsoft Learning Online Docs (Pls see "Weekly Readings" in Blackboard)	Labs 1-6 (6%) Quiz 2 (2.5%)
10/13	03 Describe core solutions and management tools on Azure Describe Azure management tools: • describe the functionality and usage of the Azure Portal, Azure PowerShell, Azure CLI, Cloud Shell, and Azure Mobile App • describe the functionality and usage of Azure Advisor • describe the functionality and usage of Azure Resource Manager (ARM) templates • describe the functionality and usage of Azure Monitor • describe the functionality and usage of Azure Service Health	Module 3 Microsoft Learning Online Docs (Pls see "Weekly Readings" in Blackboard)	Lab 7-11 (5%) Quiz 3 (2.5%)
10/20	Review / Project		
10/26	Study Week		
10/27	Study Week		
11/3	04 Describe general security and network security features Describe Azure security features: • describe basic features of Azure Security Center, including policy compliance, security alerts, secure score, and resource hygiene • describe the functionality and usage of Key Vault • describe the functionality and usage of Azure Sentinel • describe the functionality and usage of Azure Dedicated Hosts	Module 4 Microsoft Learning Online Docs (Pls see "Weekly Readings" in Blackboard)	Project 1 (15%)
11/10	04 Describe general security and network security features Describe Azure network security: • describe the concept of defense in depth • describe the functionality and usage of Network Security Groups (NSG) • describe the functionality and usage of Azure Firewall • describe the functionality and usage of Azure DDoS protection Introduction to Project 2	Module 4 Microsoft Learning Online Docs (Pls see "Weekly Readings" in Blackboard)	Lab 12-13(2%) Quiz 4(2.5%)

11/17	05 Describe identity, governance, privacy, and compliance features Describe core Azure identity services: • explain the difference between authentication and authorization • define Azure Active Directory • describe the functionality and usage of Azure Active Directory • describe the functionality and usage of Conditional Access, Multi-Factor Authentication (MFA), and Single Sign-On (SSO)	Module 5 Microsoft Learning Online Docs (Pls see "Weekly Readings" in Blackboard)	
11/24	05 Describe identity, governance, privacy, and compliance features Describe Azure governance features: • describe the functionality and usage of Role-Based Access Control (RBAC) • describe the functionality and usage of resource locks • describe the functionality and usage of tags • describe the functionality and usage of Azure Policy • describe the functionality and usage of Azure Blueprints • describe the Cloud Adoption Framework for Azure	Module 5 Microsoft Learning Online Docs (Pls see "Weekly Readings" in Blackboard)	
12/1	05 Describe identity, governance, privacy, and compliance features Describe privacy and compliance resources: • describe the Microsoft core tenets of Security, Privacy, and Compliance • describe the purpose of the Microsoft Privacy Statement, Product Terms site, and Data Protection Addendum (DPA) • describe the purpose of the Trust Center • describe the purpose of the Azure compliance documentation • describe the purpose of Azure Sovereign Regions (Azure Government cloud services and Azure China cloud services)	Module 5 Microsoft Learning Online Docs (Pls see "Weekly Readings" in Blackboard)	Labs 14-18 (5%) Quiz 5 (2.5%)
12/8	06 Describe Azure cost management and Service Level Agreements Describe methods for planning and managing costs: • identify factors that can affect costs (resource types, services, locations, ingress and egress traffic) • identify factors that can reduce costs (reserved instances, reserved capacity, hybrid use benefit, spot pricing) • describe the functionality and usage of the Pricing calculator and the Total Cost of Ownership (TCO) calculator • describe the functionality and usage of Azure Cost Management Describe Azure Service Level Agreements (SLAs) and service lifecycles: • describe the purpose of an Azure Service Level Agreement (SLA) • identify actions that can impact an SLA (i.e. Availability Zones) • describe the service lifecycle in Azure (Public Preview and General Availability)	Module 6 Microsoft Learning Online Docs (Pls see "Weekly Readings" in Blackboard)	Project 2 (15%) Lab 19-21 (2%) Quiz 6 (2.5%)
12/15	Test		Test (25%)

Missed Tests/Late Assessments

Due dates for all evaluations and assessments are posted. Evaluations can include projects, podcasts, videos, assignments, quizzes and/or tests and exams. Students are expected to meet the specified dates and deadlines. It is a best practice for all students to keep a copy of all submitted assignments.

Students who have extenuating circumstances that result in their being unable to meet the stated deadline are encouraged to contact their professor(s). A professor may (or may not) grant an extension to a posted due date. Such extension requests must be discussed prior to the due date, or very closely following. Late submission or completion of any assessments may be subject to a penalty grade deduction. Once feedback is posted and/or discussion of the assessment has taken place, students may not submit that version of the assessment for grading.

Feedback on Assessments

Feedback to students regarding graded assessments can be provided in any of the following ways: posted on LEARN@Seneca, added to Grade Centre comments, taken up synchronously, and/or discussed with students.

Students are welcome to discuss feedback on completed and submitted assessments with their professor during a synchronous class, during posted "virtual" office hours, or by a mutually agreed upon appointment.

Student Progression and Promotion Policy

Letter Grade	Percentage Grade
A+	90% to 100%
A	80% to 89%
B+	75% to 79%
B	70% to 74%
C+	65% to 69%
C	60% to 64%
D+	55% to 59%
D	50% to 54%
F	0% to 49% (Not a Pass)
OR	
EXC	Excellent
SAT	Satisfactory
UNSAT	Unsatisfactory

Listed below are a number of important links to Seneca Polytechnic policies.

- Student Progression and Promotion Policy (<http://www.senecapolytechnic.ca/about/policies/student-progression-and-promotion-policy.html>)
Outlines the requirements for students to graduate from an Ontario College Credential or an honours bachelor degree program at Seneca.
- Grading Policy (<http://www.senecapolytechnic.ca/about/policies/grading-policy.html>)
Defines the standards for assigning grades to recognize student achievement in a course and program.
- Academic Appeal Policy (<https://www.senecapolytechnic.ca/about/policies/academic-appeal-policy.html>)
Provides students with a fair, timely and consistent process to appeal decisions that impact their academic standing or progression, when the grounds for an academic appeal are met on the basis of merit of work, personal bias or unfair treatment, course management and/or extenuating circumstances.
- Prior Learning Assessment and Recognition Policy (<https://www.senecapolytechnic.ca/about/policies/prior-learning-assessment-and-recognition-policy.html>)
Establishes a framework for assessing and recognizing credit for student's prior learning.
- Transfer Credit Policy (<https://www.senecapolytechnic.ca/about/policies/transfer-credit-policy.html>)
Provides a framework for the evaluation and granting of transfer credits.

Technical Requirements

The following checklists outline the technical requirements for all students starting and continuing at Seneca:

Hardware checklist

- a computer that runs on Windows 10 or the latest Mac OSX and has up to date virus protection software
 - Windows 10 ARM64 (<https://support.microsoft.com/en-us/windows/windows-10-arm-based-pcs-faq-477f51df-2e3b-f68f-31b0-06f5e4f8ebb5>) devices are **not recommended** as they will not allow you to install AppsAnywhere, GlobalProtect, VPN, MyApps or use Virtual Commons and other virtual machine apps
- high-speed broadband access (Cable or DSL) is highly recommended. Some programs or courses require more advanced systems. Please refer to the program information page for information on specialized requirements
- headphones or speaker and a microphone for in-class conversations and meetings with your professors
- a webcam (may be required for specific courses)
- individual courses may have additional hardware requirements

Software checklist

- a web browser, such as Safari, Firefox, MS Edge, Google Chrome. Please note: You may need to upgrade your web browser to access online learning tools
- various applications are available to all full-time Seneca students, including Microsoft Office 365, Adobe Creative Suite, and Trend Micro
- Adobe Creative Suite includes a number of applications such as Premiere, Photoshop and more
- online teaching tools, including Blackboard, MS Teams, Zoom, BigBlueButton, and Webex
- individual courses may have additional software requirements for playing audio or video or other applications. You can also review the list of applications made available for home use on a Windows-based machine (<http://myapps.senecapolytechnic.ca/>)
Note: Some applications may require you to install Student VPN to access licensed software
- antimalware software must be installed on all personal devices that will be used with your Seneca account. Visit the Malware and Virus Protection (<https://students.senecapolytechnic.ca/spaces/185/it-security/wiki/view/963/malware-and-virus-protection>) page for free and paid antimalware software recommendations, or visit the Trend Micro Internet Security (<https://students.senecapolytechnic.ca/spaces/189/software/wiki/view/1360/trend-micro-internet-security>) page for a free one-year license of this commercial antimalware software

Mobile devices checklist

- Mobile devices may allow for some participation in your course(s), however they present limitations and we cannot guarantee your device will meet all your coursework needs.
- All students are required to install and use Microsoft Authenticator (<https://students.senecapolytechnic.ca/spaces/186/it-services/wiki/view/4168/microsoft-multi-factor-authentication>) to access various services at Seneca. It's an important measure that provides an added layer of security on top of the login credentials for devices. In addition to using your username and password to log into these secure services, a second factor of authentication is required so that if your password becomes compromised, the intruder will not be able to log in. Use of multi-factor authentication is currently required for Blackboard, Office 365 (<https://students.senecapolytechnic.ca/spaces/186/it-services/wiki/view/1003/office-365>) and VPN (<https://students.senecapolytechnic.ca/spaces/186/it-services/wiki/view/1024/vpn>).
- A compatible Android (<https://play.google.com/store/apps/details?id=com.azure.authenticator>) or iOS (<https://apps.apple.com/app/microsoft-authenticator/id983156458>) mobile device that can be used to install Microsoft Authenticator is required.
- A cellphone data plan is not a mandatory requirement to use the Microsoft Authenticator app. The app can be used through a Wi-Fi connection or with no data connection.
- If you have a basic cellphone, you can choose to receive an SMS or a phone call as verification for second factor authentication.
- The Microsoft Authenticator app does not store any personal data.
- Authenticating through a mobile device is the only available option.

Helpful sites to bookmark:

- MySeneca.ca (<https://outlook.office.com/mail/inbox>) – access your Seneca email account
- Learn@Seneca (<https://learn.senecapolytechnic.ca/ultra/institution-page>) – Seneca's learning management system and intranet portal

Seneca Polytechnic Library Resources

Be sure to begin all research, assignment support and career preparation at Seneca Polytechnic Libraries (<http://library.senecapolytechnic.ca>) website. Students can find information about our services and collections including, print and e-books, databases that will lead to thousands of articles in magazines, newspapers, journals, encyclopedias, carefully selected websites, how-to tutorials, streamed videos and much more.

Citation Style Guidelines (<https://library.senecapolytechnic.ca/citingsources>): APA/MLA. Please check with your professor on the preferred formatting.

Seneca Policies

Below are some Seneca policies and links to more information.

Academic Integrity Policy (<https://www.senecapolytechnic.ca/about/policies/academic-integrity-policy.html>)

Seneca upholds a learning community that values academic integrity, honesty, fairness, trust, respect, responsibility and courage. These values enhance Seneca's commitment to deliver high-quality education and teaching excellence, while supporting a positive, equitable and inclusive learning environment. Review section 3 of the policy for details regarding approaches to supporting integrity. Section 4 and Appendix B of the policy describe various sanctions that can be applied, if there is suspected academic misconduct (e.g., contract cheating, cheating, falsification, impersonation or plagiarism).

Please visit the Academic Integrity at Seneca (<http://open2.senecac.on.ca/sites/academic-integrity/for-students>) website to understand and learn more about how to prepare and submit work so that it supports academic integrity, and to avoid academic misconduct.

Accessible Learning Services Policy (<https://www.senecapolytechnic.ca/about/policies/accessible-learning-services-policy.html>)

Seneca provides reasonable accommodations to students with disabilities to support academic success. If you require accommodation, contact the Accessible Learning Services Office (senecacnas@senecapolytechnic.ca) to initiate the process for documenting, assessing and implementing your individual accommodation supports for the classroom and Work-Integrated Learning (WIL) environments.

Accommodated students are required to meet the expected learning outcomes of courses. Accommodations do not surpass the need for safety or supersede academic policies and requirements.

Generative Artificial Intelligence (GenAI) Policy (<https://www.senecapolytechnic.ca/about/policies/generative-ai-policy.html>)

Seneca is committed to leveraging available technologies to work smarter, better and more efficiently. This includes the use of Generative Artificial Intelligence (GenAI) technologies. GenAI Frequently Asked Questions are available for students (<https://students.senecapolytechnic.ca/spaces/288/copilot-seneca/wiki/view/15227/frequently-asked-questions-faqs>). Violations of this policy may be subject to discipline.

Inclusion Policy (<https://www.senecapolytechnic.ca/about/policies/inclusion-policy.html>)

Every Seneca community member has the right to study and work in an environment that is free from discrimination, harassment, hate and racism. Individuals found to exhibit prohibited acts may face sanctions in accordance with relevant Seneca policies. Information and assistance is available by contacting the Student Conduct Office at student.conduct@senecapolytechnic.ca.

Recording Lectures and Educational Activities Policy (<https://www.senecapolytechnic.ca/about/policies/recording-lectures-and-educational-activities-policy.html>)

Seneca students and employees are required to follow the framework of this policy for the collection, use, disclosure, and disposal of recordings of educational activities. The policy reinforces Seneca's support of recordings as a means to promote student learning and success.

By attending synchronous classes, students are consenting to the collection and use of their personal information for the purposes of administering the class and associated coursework. To learn more about Seneca's privacy practices, review Seneca's Privacy Notice (<https://www.senecapolytechnic.ca/privacy.html>).

Student Assessment Policy (<https://www.senecapolytechnic.ca/about/policies/student-assessment-policy.html>)

Seneca's approach to student assessment is grounded in commitments to equity and accessibility for students. Student work in academic courses will be assessed and evaluated to provide ongoing and meaningful feedback about their progress and development.